Two-dimensional powder diffraction in layered van der Waals films: quantitative X-ray area-detector modeling

1 Introduction

For layered thin films, the central structural question is often not only which phase is present, but how the crystallites are oriented relative to the substrate. In layered van der Waals solids and related two-dimensional materials, this orientational state can directly control the properties that motivate the films in the first place.[1–3] In layered halides such as PbI_2, transport and device response can depend strongly on whether the layers lie parallel to the substrate.[4] In topological bismuth chalcogenides, strong c-axis alignment is needed to preserve the desired electronic response.[5] In transition-metal dichalcogenides, catalytic activity can increase when platelet edges are exposed by the film orientation.[6] Comparable orientational states also occur across a wider set of layered thin-film families, including telluride chalcogenides (e.g., Bi_2Te_3, Sb_2Te_3), dichalcogenides (e.g., MoS_2, NbS_2), layered halides and oxyhalides (e.g., BiI_3, BiOI), and laminated 2D-flake films such as $Ti_3C_2T_x$ MXenes and graphene-oxide papers.[7–14]

Wide-angle x-ray scattering (WAXS) with an area detector is attractive for this problem because a single exposure probes a large reciprocal-space volume rich with information of both structure and orientation.[15–19] The difficulty is that many layered films do not fall cleanly into either of the standard diffraction limits. They are neither single crystals, where reciprocal space consists of discrete points, nor fully isotropic three-dimensional (3D) powders, where orientational averaging produces spherical shells.[20] Instead, a common thin-film orientational state is the two-dimensional (2D) oriented powder, in which plate-like crystallites share an approximately aligned layer normal while remaining azimuthally randomized in-plane.[3, 20, 21] This intermediate orientational state produces detector patterns that inherit features of both limiting cases while necessitating nontraditional analysis techniques.[20]

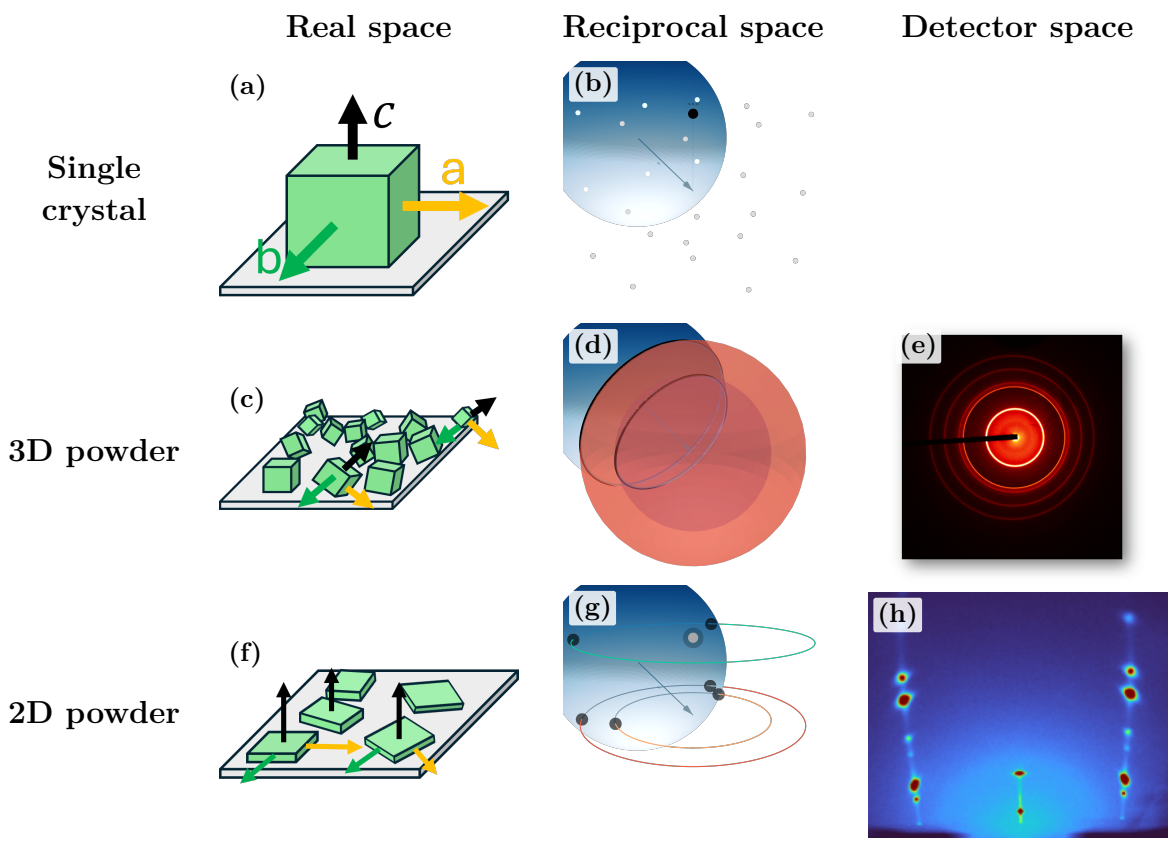

Figure 1: Diffraction across three orientational limits: single crystal, 3D powder, and 2D powder. Columns show the corresponding real-space, reciprocal-space, and detector-space views, while rows group the orientational limits. Panels (a,b) show a single crystal in real space and reciprocal space. Panels (c–e) show a 3D powder, with randomly oriented crystallites, spherical reciprocal-space shells, and Debye–Scherrer rings on the detector. Panels (f–h) show a 2D powder, with aligned surface normals and in-plane orientational disorder, reciprocal-space rings with specular Bragg features, and the corresponding detector pattern with symmetry about the central axis.

Figure 1 summarizes the real-space, reciprocal-space, and detector-space consequences of these three orientational limits. In the Ewald construction, elastic scattering requires that reciprocal-space intensity lie on the Ewald sphere of radius $|\mathbf{k}|$.[22] For a single crystal, reciprocal space consists of isolated Bragg points (Fig. 1b), so at fixed incidence angle the Ewald sphere typically intersects only a small subset of reflections and the detector records sparse spots.[22] For a 3D powder, random crystallite orientations smear the intensity of each Bragg point over a spherical shell, and the Ewald-sphere intersection produces Debye–Scherrer rings on the detector (Fig. 1d, Fig. 1e).[22] For a 2D powder, azimuthal disorder about the common c axis smears each off-specular Bragg point into an in-plane ring about Q_z, while specular $(00L)$ reflections remain localized near the Q_z axis (Fig. 1g).[15, 20] The corresponding illustration in Fig. 1f schematically shows 2D platelets with aligned growth, and a representative 2D-powder detector pattern is shown in Fig. 1h. When the Ewald sphere intersects this reciprocal-space distribution, the detector records hybrid signatures: pairs of symmetric spots from the off-specular ring intersections and single-crystal-like features near

the specular stripe from $(00L)$ intensity.[15, 17, 23] Mosaicity then redistributes intensity, broadening the ring intersections into arcs and spreading the $(00L)$ features into cap-like intensity near the specular region.[15, 20]

This hybrid detector-space information is what makes the 2D-powder problem both rich and difficult. Many standard WAXS workflows reduce the area-detector image to 1D profiles by azimuthal integration, often with masking near the direct beam and specular region.[17, 19, 24–26] Those steps are well suited to nearly isotropic powders, but for 2D powders they deweight or remove the arcs and specular caps that are most sensitive to out-of-plane correlations, stacking, and mosaic tails.[15, 17] Stacking disorder is also commonly inferred from 1D interference models and refinements,[27–29] which do not directly exploit the full two-dimensional anisotropy of oriented-film scattering. More general intermediate orientation distributions can be treated with orientation-distribution-function (ODF) refinements or preferred-orientation corrections.[30–35] Those approaches are more flexible, but they are also more general than is needed for the narrowly constrained 2D-powder case considered here.[34, 35] Recent radial-profile approaches have made thin-film GIXD substantially more quantitative with extracted one-dimensional profiles but peak overlap and background separation can remain limiting in strongly anisotropic patterns, especially at small incidence angles.[36] For the layered films considered here, that level of generality is often unnecessary. The morphology and the measured patterns already motivate an axial symmetry about the film normal together with a low-dimensional out-of-plane mosaicity kernel.[3, 20, 21] We therefore assume this physically motivated 2D-powder orientation distribution a priori and refine its parameters by forward-modeling the full 2D detector image, so that the arc and specular (Bragg-rod) line shapes in Fig. 1 act as direct constraints while reducing computational cost and improving fit precision.[15, 17]

Here we present a quantitative forward model for WAXS from two-dimensional powders that predicts full area-detector images and then refines stage-specific detector-derived observables against measured data. Prior work has established the main ingredients needed for such a framework, including grazing-incidence scattering formalisms,[37, 38] reciprocal-space mapping and indexing for oriented thin films,[15, 17, 19, 23] and forward simulation of diffraction from textured films.[39] Complementary atomistic tools have also been developed for constructing commensurate multilayer 2D heterostructure supercells and visualizing their reciprocal-space diffraction signatures, including Nookiin.[40] In contrast, the present work takes the crystallographic model as input and carries the calculation in detector space through the coupled effects of detector geometry, grazing-incidence propagation, mosaicity, and stacking disorder. Geometry is constrained by detector-space peak positions, mosaicity by local profile shapes, ordered structure by selected Bragg-ROI intensities, and disorder by selected-rod profiles. Our contribution is to combine these ingredients in a single self-consistent refinement scheme tailored to 2D powders. We validate the ordered 2D-powder limit using Bi_2Se_3 and Bi_2Te_3 films,[5, 7] and we then apply the same framework to disordered PbI_2 to extract effective stacking-disorder parameters from diffuse Q_z intensity in selected Q_r/Q_z rod profiles that are typically masked in 1D-integrated analyses (see sup-

porting information, Fig. S1, Sec. S1.3, and Secs. S2–S3).[28, 29] By embedding geometric and sample effects within a single forward model, we avoid separating geometric corrections from structural fitting and instead refine geometry, mosaicity, ordered intensity, and disorder within a common detector-space framework.

2 Diffraction Model

We model the measured area-detector intensity by building the diffraction process in the same order used throughout this section. We first determine the allowed exit vectors $\mathbf{k}_f$ for the simplest elastic-scattering geometry, then extend that construction to the finite beam, sample, and detector geometry of the experiment. For each allowed reciprocal-space intersection, the crystallographic model supplies the structure-factor weight, and the mosaicity model redistributes that weight over the nearby $\mathbf{k}_f$ values appropriate to a two-dimensional powder. The result is a full detector-space image calculation with intensity deposited on detector pixels. That image is the common forward object for the workflow, but the refinement objectives are stage-specific: peak centers for geometry, local Gaussian-plus-Lorentzian line shapes for mosaicity, detector-space Bragg ROI integrals for ordered intensity, and selected-rod profiles for disorder.

2.1 Diffraction Geometry

We first consider a single incident ray striking a perfectly aligned crystallite, with no beam divergence, no refraction, and no absorption. Elastic scattering requires

$$|\mathbf{k}_i| = |\mathbf{k}_f| = \frac{2\pi}{\lambda}, \qquad \mathbf{Q} = \mathbf{k}_f - \mathbf{k}_i.$$

Diffraction occurs when the momentum transfer matches a reciprocal-lattice vector, $\mathbf{Q} = \mathbf{G}$. In the Ewald construction, this means that the reciprocal-space point for that reflection lies on the Ewald sphere of radius $|\mathbf{k}_i|$. A useful reciprocal-space visualization of this construction is shown in Fig. 1b. The simplest diffracted ray is obtained directly from

$$\mathbf{k}_f = \mathbf{k}_i + \mathbf{G}.$$

For a fixed $|\mathbf{G}|$, the set of possible reflection orientations forms a Bragg sphere of radius $|\mathbf{G}|$ about the reciprocal-space origin. The allowed exit vectors are found by intersecting that Bragg sphere with the Ewald sphere, as visualized by the spherical reciprocal-space shell in Fig. 1d, and then propagating the resulting exit ray to the detector.

For a single $\mathbf{k}_f$, we propagate that ray from the sample to the detector plane. The 2D area detector is positioned at a distance D from the sample along the lab y axis, with tilts about the x and z axes denoted β and γ, respectively, as shown in Fig. 2. An additional in-plane detector rotation χ accounts for rotation of the detector axes within the detector plane. A diffracted ray with exit wavevector $\mathbf{k}_f$ in the lab frame intersects the detector at a scattering angle 2θ, defined as the angle between $\mathbf{k}_f$ and the y axis, and at a detector-plane azimuth ϕ, measured from the detector x' axis.

Recreating a real measurement requires a family of trial incident rays and sample configurations. In the lab frame, the nominal incident wavevector $\mathbf{k}_i$ propagates along the y

axis, but different trial rays can arrive with different transverse positions and directions, characterized by widths w_x, w_z and rms divergences $\Delta\theta_x, \Delta\theta_z$. Numerically, we sample these trial rays and propagate each one to the sample.

The goniometer defines a rotation axis about which the sample normal rotates, as shown in Fig. 3. In the aligned configuration, a rectangular film of in-plane dimensions W and H lies in the x-y plane with thickness t along the sample normal. The rotation axis lies within this plane, with the incident beam parallel to the x-y plane and perpendicular to the rotation axis. In practice, the beam center and sample center may be displaced from the goniometer axis by vertical offsets z_B and z_S, respectively. Rotating the sample about this axis sets the incidence angle θ_i between the incident wavevector $\mathbf{k}_i$ and the sample surface, as shown in Fig. 2. If the substrate is not perfectly parallel to the x-y plane, the sample normal can also be tilted by a residual angle δ. For each trial ray, we compute its intersection with the sample plane. The finite beam size at grazing incidence produces an elongated footprint of length X_s (Fig. 3). Rays that fall outside the finite illuminated window set by W and H miss the sample and are discarded. In this way, the model explicitly captures finite-footprint illumination and footprint truncation.

For rays that intersect the film, refraction also changes the ray geometry itself. We model the film by a complex refractive index $\tilde{n} = 1 - \delta_n + i\beta_n$ and compute transmitted internal wavevectors using Snell's law together with continuity of the in-plane components. The geometric task remains the same as in the simplest case: for each trial incident ray and sample configuration, determine the corresponding allowed exit vector or vectors $\mathbf{k}_f$ and propagate them to the detector.

After the allowed $\mathbf{k}_f$ vectors are generated, each one is assigned a multiplicative weight. Beam-distribution weights arise first from the sampled incident-ray phase space, reflecting the intensity associated with each beam position and direction. Geometric weights then arise from whether the incident ray reaches the finite illuminated sample window. Optical weights arise from Fresnel transmission at the surface and from absorption along the incident and exit paths through the film thickness t.

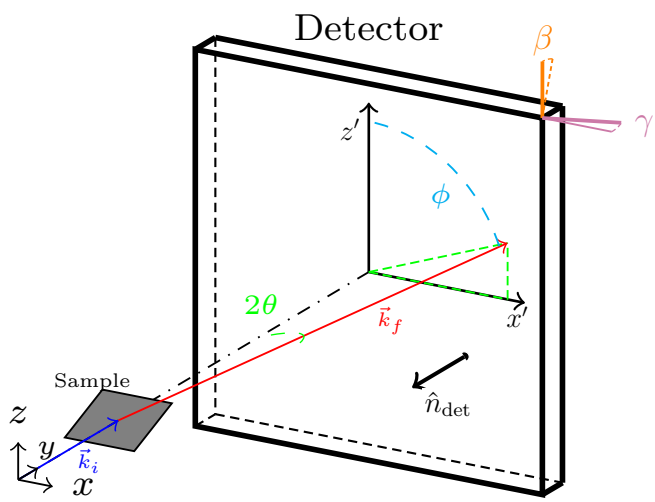

Figure 2: Overall lab-frame geometry. The incident wavevector $\mathbf{k}_i$ strikes the sample, and a representative exit wavevector $\mathbf{k}_f$ reaches the detector at scattering angle 2θ and detector-plane azimuth ϕ. The detector at distance D is tilted by β and γ.

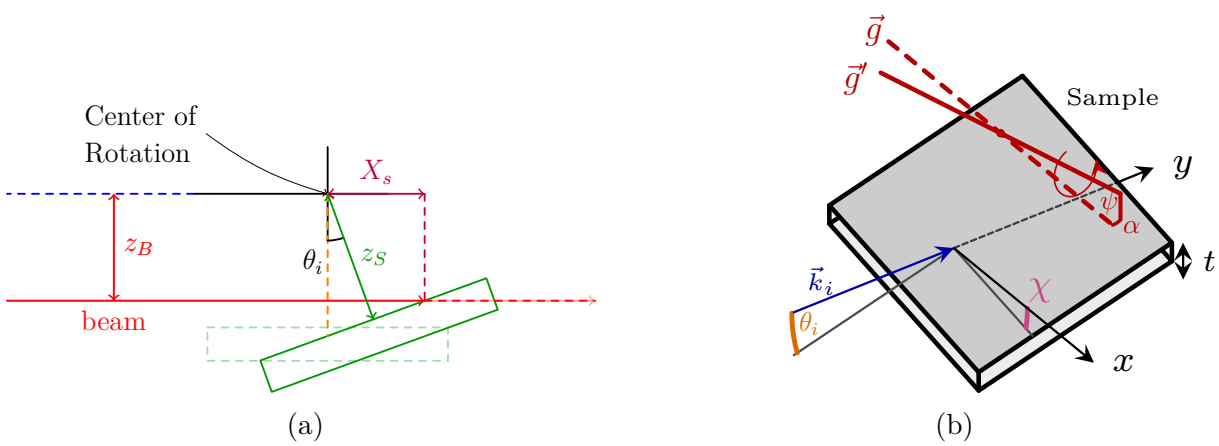

Figure 3: Two-stage sample geometry. Left: rotation about the goniometer center with sample offset z_S and beam offset z_B. θ_i is the incidence angle. Right: full alignment adds tilt δ and footprint length X_s within an illuminated window (W, H) on a film of thickness t. $\vec{g}$ is the goniometer arm (normal to $\hat{y}$), and $\vec{g}'$ is the misaligned arm obtained by rotations α and ψ.

2.2 Structure Factor

We enumerate all Miller triplets (h, k, l) whose scattering falls within the detector acceptance, construct the corresponding reciprocal-lattice vector $\mathbf{G}$, and evaluate the momentum-transfer magnitude $Q = |\mathbf{Q}| = |\mathbf{G}|$ for that reflection. For each allowed intersection, the geometry step supplies the associated set of $\mathbf{k}_f$ vectors on the detector, and the crystal structure contributes a multiplicative factor obtained by evaluating the structure-factor function at

that reciprocal-space point. In the discrete Bragg case considered here, that weight is written as $I_{0,\mathbf{G}} \propto |F_{\mathbf{G}}|^2$, where $\mathbf{Q} = \mathbf{G}$ at the intersection. More generally, the same operation can be viewed as sampling a function $F(\mathbf{Q})$ at the intersected reciprocal-space coordinate, so the crystallographic weight may be defined on either integer-indexed reciprocal-lattice points or a continuous reciprocal-space variable.

Conventional powder calculations often replace symmetry-equivalent reflections by a multiplicity factor. We do not do that here. In a two-dimensional powder, the measured intensity of symmetry-related reflections depends on the experimental geometry and on the crystallite orientation distribution, so the effective multiplicity is not a single geometry-independent number. We therefore retain every allowed $\mathbf{G}$ explicitly, evaluate its structure factor at the appropriate Q, and let multiplicity and orientation effects emerge within the forward model.

For each reflection with reciprocal-lattice vector

$$\mathbf{G} = h\mathbf{a}^* + k\mathbf{b}^* + l\mathbf{c}^*,$$

we compute the structure factor as

$$F_{\mathbf{G}} = \sum_a o_a \, f_a(Q) \, \exp[2\pi i \, \mathbf{G} \cdot \mathbf{r}_a] \, \exp\Big[-2\pi^2 \, \mathbf{G} \cdot \mathbf{U}_a \cdot \mathbf{G}\Big] \,, \tag{1}$$

where the sum runs over atoms a in the unit cell, o_a is the site occupancy, $f_a(Q)$ is the atomic form factor evaluated at the momentum-transfer magnitude of that reflection, and $\mathbf{r}_a = (x_a, y_a, z_a)$ gives the fractional atomic position generated by the space-group symmetry (Wyckoff site), so that $\mathbf{G} \cdot \mathbf{r}_a \equiv hx_a + ky_a + lz_a$. If we refine positions, we vary only the free parameters of the occupied Wyckoff sites, not independent per-atom shifts. The Debye–Waller factor is written in anisotropic tensor form, where $\mathbf{U}_a$ is the mean-square displacement tensor (ADP) for atom a, expressed in the same basis convention used for $\mathbf{G}$. Operationally, once an allowed $\mathbf{k}_f$ is found, the corresponding ray weight is multiplied by the crystallographic factor $|F|^2$ evaluated at that reciprocal-space intersection. In refinement, $\mathbf{U}_a$ may be constrained by symmetry or reduced parameterizations when warranted by the data.

2.3 Mosaicity

The mosaic picture is most easily understood by comparing the three orientational limits introduced in Fig. 1b, Fig. 1d, and Fig. 1g. For a perfect single crystal (Fig. 1b), a reflection carries its full weight at a single Bragg spot. For a 3D powder (Fig. 1d), random crystallite orientations spread that weight uniformly over the Bragg sphere, so the Ewald-sphere intersection gives a circle of equivalent reciprocal-space points and associated exit vectors $\mathbf{k}_f$. A 2D powder (Fig. 1g) lies between these limits: the allowed $\mathbf{k}_f$ values still lie on that circle, but out-of-plane mosaicity makes some portions of the circle more probable than others. Mosaicity is therefore treated as a redistribution of the ideal per-reflection intensity over

the allowed $\mathbf{k}_f$ family, rather than as a change to the underlying diffraction condition or structure factor.

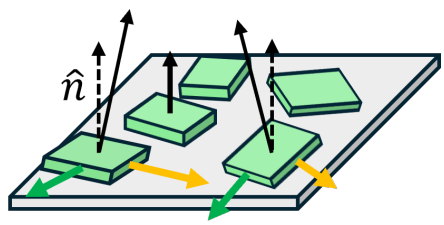

Figure 4: Schematic of crystallite layer-normal misorientation (mosaicity) about the surface normal. A distribution of crystallite orientations may form in a thin film.

For a reflection, we work in the (G_r, G_z) cross-section of reciprocal space, where G_r is the projection of the reciprocal lattice vector onto the plane of the substrate, and G_z is the component along the surface normal. In this section, the perfect-crystal Bragg spot appears as a point on the Bragg circle. In the full three-dimensional picture, random in-plane rotations generate the circle of equivalent reciprocal-space intersections and associated $\mathbf{k}_f$ values described above, consistent with the reciprocal-space evolution from Fig. 1b to Fig. 1d to Fig. 1g. Out-of-plane misorientation then replaces the uniform weighting of that ideal circle by a line density $I_0\omega(\Delta\theta)$, where I_0 is the total intensity associated with the reflection before mosaic redistribution.

We describe this out-of-plane mosaicity by a misorientation density $\omega(\Delta\theta)$, where $\Delta\theta$ is the crystallite tilt angle relative to the substrate normal. The implemented density is a normalized Gaussian-plus-Lorentzian angular distribution, evaluated in wrapped form on $(-\pi, \pi]$ to ensure the probability is conserved on one full angular period and avoids discontinuity at $\Delta\theta = \pm\pi$. The Gaussian and Lorentzian widths are independent, and the Lorentzian component is necessary to capture the long tails needed for off-condition specular intensity. The explicit wrapped Gaussian and wrapped Lorentzian forms are given in the supporting information, Sec. S1.1. We report the Lorentzian full width at half maximum, the Gaussian full width at half maximum, and the relative Lorentzian weight.

These components are related to the full reciprocal lattice vector $\boldsymbol{G} = (H, K, L)$, but the relationship depends on the crystal symmetry. For a general crystal, G_r is defined as the magnitude of the projection of $\boldsymbol{G}$ onto the plane, and G_z is the component along the normal direction. The mixed Gaussian-plus-Lorentzian form is motivated by how different regions of the detector image sample ω. Off-specular reflections with $G_r \neq 0$ are spread by random in-plane rotation around an azimuthal ring, so their observed widths are governed mainly by the core of the misorientation distribution. Specular $(00L)$ reflections, however, remain concentrated near the Q_z axis rather than being diluted around that ring. Consequently, even weak tail probability can remain visible in the specular region, allowing intensity from misoriented crystallites to appear at a nominal θ_i where a perfectly aligned crystallite would be off condition. The Lorentzian component is therefore introduced to capture the long

tails needed to reproduce the persistence of multiple specular reflections at a single nominal incident condition, beyond what beam divergence alone would provide.

Figure 5 summarizes how this reciprocal-space broadening appears in the $\{G_r, G_z\}$ plane and how it maps onto characteristic detector-space features for in-plane and specular conditions.

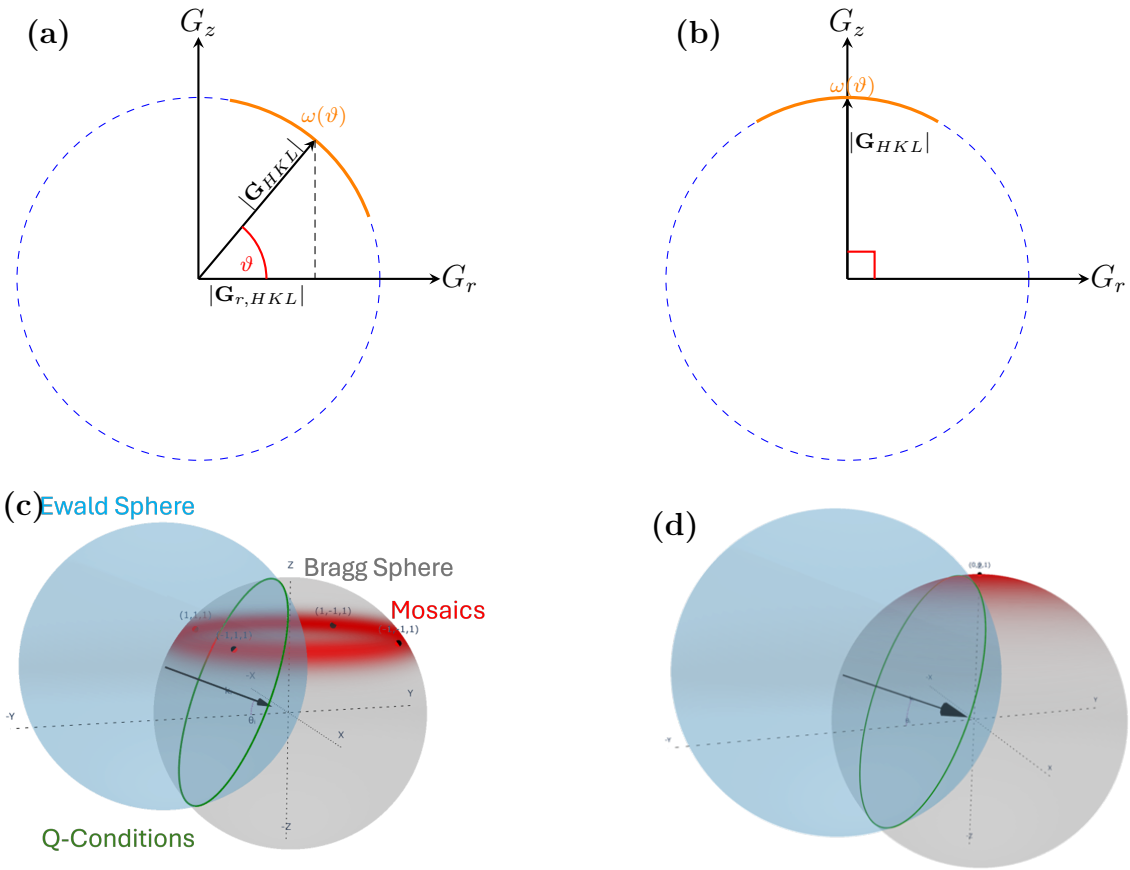

Figure 5: Mosaic broadening viewed in the $\{G_r, G_z\}$ plane (a,b) and its detector-space manifestation (c,d). Intensity is redistributed along the Bragg circle with line density $I_0\,\omega(\Delta\theta)$. The in-plane condition with $G_r \neq 0$ (a,c) yields two arcs of reflection, while the specular reflection with $G_r = 0$ (b,d) concentrates intensity near the polar cap, forming a single arc.

Random in-plane rotations about the film normal revolve that line density about the G_z axis, generating an axially symmetric distribution on the Bragg sphere. Let ϑ denote the polar angle on that sphere, and let ϑ_B denote the Bragg position for the reflection, so that the mosaic misorientation is $\Delta\theta = \vartheta - \vartheta_B$.

We normalize the corresponding Bragg-sphere surface density $\rho(\vartheta)$ so that $\int \rho(\vartheta)\, dA = I_0$, yielding

$$\rho(\vartheta) = \frac{I_0}{2\pi G^2 \sin\vartheta}\,\omega(\vartheta - \vartheta_B) \qquad 0 < \vartheta < \pi, \tag{2}$$

where $G \equiv |\mathbf{G}|$ is the magnitude of the reciprocal-lattice vector, i.e. the Bragg-sphere radius for the reflection. The factor $1/\sin\vartheta$ is the Jacobian for converting the one-dimensional

tilt distribution ω into a surface density. This surface density supplies the weights used to distribute the ideal intensity I_0 over the neighboring $\mathbf{k}_f$ rays generated by misoriented crystallites.

The reciprocal-space consequences appear on an area detector in Fig. 6. Strong out-of-plane mosaicity concentrates specular intensity near discrete G_z values at $G_r = 0$ (region A), while random in-plane orientation distributes in-plane reflections uniformly in azimuth about G_z, producing ring-like features (region B).

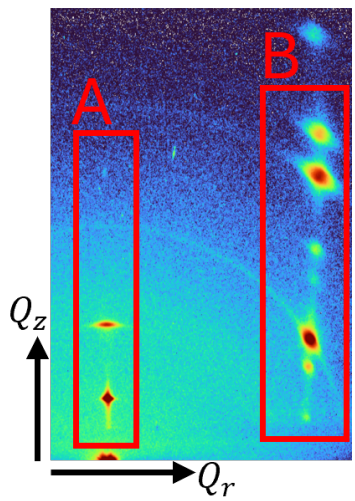

Figure 6: Representative log-scale WAXS pattern from a layered 2D-powder film. Region A: multiple specular reflections are visible, although the idealized picture would suggest only one; the lowest-Q_z specular peak exhibits a star-like shape. Region B: in-plane scattering.

In the in-plane region (Fig. 6B), the Ewald sphere intersects the broadened Bragg-sphere feature along an extended band. Azimuthal averaging then produces band-like scattering whose radial width and line shape are set primarily by the Gaussian core of ω and by the Jacobian factor $1/\sin\vartheta$ in Eq. (2). In the specular region (Fig. 6A), the same mosaic spread produces cap-like broadening about the specular reflections. More than one specular peak is visible in region A, even though the idealized specular picture would suggest only a single peak, and the lowest-Q_z specular peak in particular exhibits a star-like profile rather than purely isotropic broadening. This shape is attributed to the finite effective thickness of the Ewald sphere, set by beam divergence and related resolution effects, which becomes comparatively more important for smaller reciprocal-lattice magnitudes G. In addition, small rotations of the Ewald sphere produce smaller positional redistributions closer to the origin, so the broadening is concentrated mainly along q_z rather than appearing as strong azimuthal spreading. These off-condition specular contributions arise from crystallites whose layer normals are tilted by $\Delta\theta$ away from the mean. Because the redistributed intensity remains concentrated near the Q_z axis, the specular region is much more sensitive than the in-plane arcs to the tails of ω; it therefore provides the primary constraint on the Lorentzian

component, with only weak additional variation from the geometric factor in Eq. (2).

3 Refinement workflow

Because the forward model combines beam phase space, detector geometry, sample alignment, mosaicity, and crystallographic intensity within a single image calculation, a fully global refinement from arbitrary starting values would lead to strong correlations between instrumental and sample-specific parameters. We therefore use a staged refinement strategy. Each stage uses the detector-space observable that is most diagnostic for that parameter class before the next class of parameters is introduced. Beam and detector terms first establish the experimental geometry shared across measurements. Sample alignment then determines where reflections appear on the detector. With the reflection positions established, mosaicity is constrained by local line shape, ordered structure by selected relative Bragg intensities, and stacking disorder by selected Q_z rod profiles. The workflow is summarized in Table 1.

Table 1: Staged refinement workflow. All sample images are dark-subtracted before the listed observables are formed.

Step	Stage	Data	Parameters refined	Primary information
1	Beam characterization	Direct-beam images at several detector distances	Beam widths w_x, w_z and divergences $\Delta\theta_x$, $\Delta\theta_z$	Incident phase space
2	Detector geometry calibration	Powder calibrant image	Detector distance D, detector tilts (β, γ), and beam center (x_0, y_0)	Detector mapping
3	Sample alignment and goniometer misalignment	Indexed reflection centers from sample images	Sample geometry $(\theta_i, \delta, z_B, z_S)$, in-plane rotation χ, goniometer misalignment (α, ψ)	Detector-space peak positions
4	Mosaicity/profile refinement	Selected local detector ROIs and local $I(\phi)$ profiles	Gaussian-plus-Lorentzian mosaic parameters	Center-aligned profile shapes and specular tails
5	Ordered-structure refinement	Selected detector-space Bragg ROIs from multiple dark-subtracted images	Image-level scales and ordered-structure parameters entering Eq. (1), including occupancies o_a, atomic positions $\mathbf{r}_a$, static displacements $\Delta\mathbf{r}_a$, and anisotropic Debye–Waller terms $\mathbf{U}$	Relative ROI-integrated Bragg intensities
6	Disorder refinement	Selected Q_r/Q_z rod profiles from caked signal and normalization fields	Image-level scales, stacking-fault or slip probability, optional finite-stack terms, and polytype-mixture terms	Diffuse Q_z redistribution along the selected rod

The first two stages isolate the instrumental contribution to the measured image. Direct-beam images collected at several detector distances determine the incident-beam widths and divergences used in the forward model. These quantities define the baseline phase-space spread of the incoming beam and therefore the instrumental broadening that would be present even for an ideal sample. Detector distance, tilts, and beam center are then refined from hBN Debye–Scherrer rings. This calibration fixes the mapping from reciprocal-space intersections to detector coordinates independently of the sample, so later discrepancies in peak position or line shape are not absorbed spuriously into sample parameters.

With the instrumental terms fixed, refinement proceeds to the sample geometry. For

each sample image, alignment and goniometer misalignment are determined from indexed Bragg-peak positions rather than from global intensities, because feature location is the most direct signature of geometric error. Reflections spanning both the specular and off-specular regions are included so the fit is constrained by the central Q_z-dominated features and by the off-axis intersections of the reciprocal-space rings. This step establishes the actual incidence condition and illuminated geometry of the experiment, which must be known before peak widths and intensities can be interpreted physically.

Mosaicity is refined only after the reflection positions have been fixed, so the remaining variation in line shape can be attributed to crystallite misorientation rather than to geometric offset. The fit uses selected local detector regions around indexed reflections and compares measured and simulated local profiles after center alignment. When multiple incidence settings are available, they share the same Gaussian-plus-Lorentzian mosaic density. The Gaussian and Lorentzian components have independent widths and together describe a wrapped angular distribution. Off-specular arcs constrain the Gaussian-dominated core width, whereas the persistence and decay of specular intensity constrain the Lorentzian tail more strongly.

Only after the geometric and mosaic terms have converged are the remaining Bragg-intensity variations assigned to ordered structure. The ordered stage uses background-subtracted integrated counts from selected detector-space Bragg ROIs, compared with the simulated detector-carried intensity deposited in those same ROIs. One scale is allowed for each image to absorb exposure and flux differences, while the shared ordered-structure parameters control the reflection-to-reflection intensity ratios.

Disorder parameters are introduced only after the ordered model has been pushed as far as it can go. At that point the beam, detector, sample geometry, mosaic/profile state, lattice parameters, and ordered intensity baseline are held fixed. Scale factors remain image-level terms: they are shared across the fitted observables from a given image but are not constrained to be identical between different images. The final observable is a selected Q_r/Q_z rod profile formed from caked signal and normalization fields by summing signal and normalization over the selected rod mask before division. The disorder stage changes the stacking-disorder parameters and any image-level scale terms, rather than introducing independent peak areas. This separation is essential because diffuse intensity along Q_z near and between Bragg peaks is otherwise easily confounded with average-structure or broadening effects.

4 Results: ordered films

We first applied the staged workflow to ordered MBE-grown Bi_2Te_3 and Bi_2Se_3 films. The detector images contained the specular features and off-specular arcs expected for a 2D oriented powder with little stacking disorder. These two materials therefore provided the ordered-film benchmark.

Placeholder figure

`ordered_detector_summary.png`

Panels: (a) Bi_2Te_3 detector comparison; (b) Bi_2Se_3 detector comparison; (c) representative local profiles; (d) post-fit Q_z intensity checks.

Figure 7: Ordered-film detector-space fits. (a) Bi_2Te_3 measured, simulated, and residual detector views after the final staged refinement. (b) Corresponding Bi_2Se_3 detector comparison. (c) Representative local profiles from specular and off-specular detector ROIs used to check the mosaic/profile state after center alignment. (d) Post-fit Q_z intensity checks for ROI-integrated Bragg intensities in the two ordered films. The comparison is made after the detector-space geometry and local line-shape terms have been fixed.

Figure 7 summarizes the final ordered-film fits in the same order as the refinement workflow. The measured, simulated, and residual detector panels show that the model placed the specular peaks and off-specular arcs at the observed detector positions for both Bi_2Te_3 and Bi_2Se_3. This is the most sensitive check of the beam-center, detector-distance, detector-tilt, and sample-alignment parameters, because even small geometric errors would produce peak-position offsets across the image.

The local profile checks then isolate line-shape information from integrated intensity information. The specular-cap and off-specular-arc profiles were reproduced by the same Gaussian-plus-Lorentzian mosaic profile. The final Q_z checks compared the ROI-integrated Bragg intensities after the geometry and line-shape parameters had been fixed. Agreement in this view indicated that the ordered structure-factor description was sufficient for these films within the resolution of the present workflow. The important result is that the same refinement remained stable in both systems, thereby setting the baseline for the PbI_2 disorder analysis.

5 Results: diffuse scattering in PbI_2

We next applied the workflow to CVD-grown PbI_2 films. PbI_2 was the disorder-sensitive case because the measured image contained diffuse intensity along selected Q_z rods that was not captured by the ordered baseline alone. The analysis began in the same way as for the ordered films: beam and detector terms were fixed, sample alignment was determined from detector-space feature positions, mosaic/profile parameters were constrained from local line shapes, and the ordered PbI_2 intensity model was fixed from the Bragg peak positions and areas. The stacking-fault model was introduced only after that ordered baseline was established.

The final selected-rod intensity was represented as

$$I_{hk}(L) = \mathcal{A} F_{hk}^2(L) R_{hk}(L; p, N, \ldots), \tag{3}$$

where $\mathcal{A}$ is the image-level scale factor, $F_{hk}^2(L)$ is the ordered PbI_2 rod intensity, R_{hk} is the analytical stacking-fault correction, p is the fitted slip or fault parameter, and N is an effective finite-stack number. When multiple images are fit, $\mathcal{A}$ is shared within each image but is not constrained to be identical across images. The disorder observable was built from the scattering magnitude and from the polar angle η of the scattering vector relative to the film normal,

$$Q = \frac{4\pi}{\lambda} \sin\left(\frac{2\theta}{2}\right), \qquad Q_r = |\sin\eta| Q, \qquad Q_z = \cos\eta\, Q. \tag{4}$$

This is the same reciprocal-space transformation used in the caked selected-rod analysis; η denotes the polar angle used for the Q_r/Q_z conversion and is distinct from the detector-plane azimuth ϕ used in the detector-geometry discussion. A rod mask kept bins satisfying $|Q_r - Q_{r,0}| \leq \Delta Q_r$ and the chosen Q_z bounds, with one detector branch used for the default fit and the opposite branch held as a validation profile.

The selected-rod profile was fit directly. Beam, detector geometry, sample alignment, mosaic/profile state, lattice constants, and the ordered PbI_2 baseline were held fixed. Known Bragg positions on the retained branch defined the profile locations, and the number of film layers was used as an input to the stacking model. The free disorder terms were the stacking-fault slip probability, p, and the relative ratio of the polytypes present.

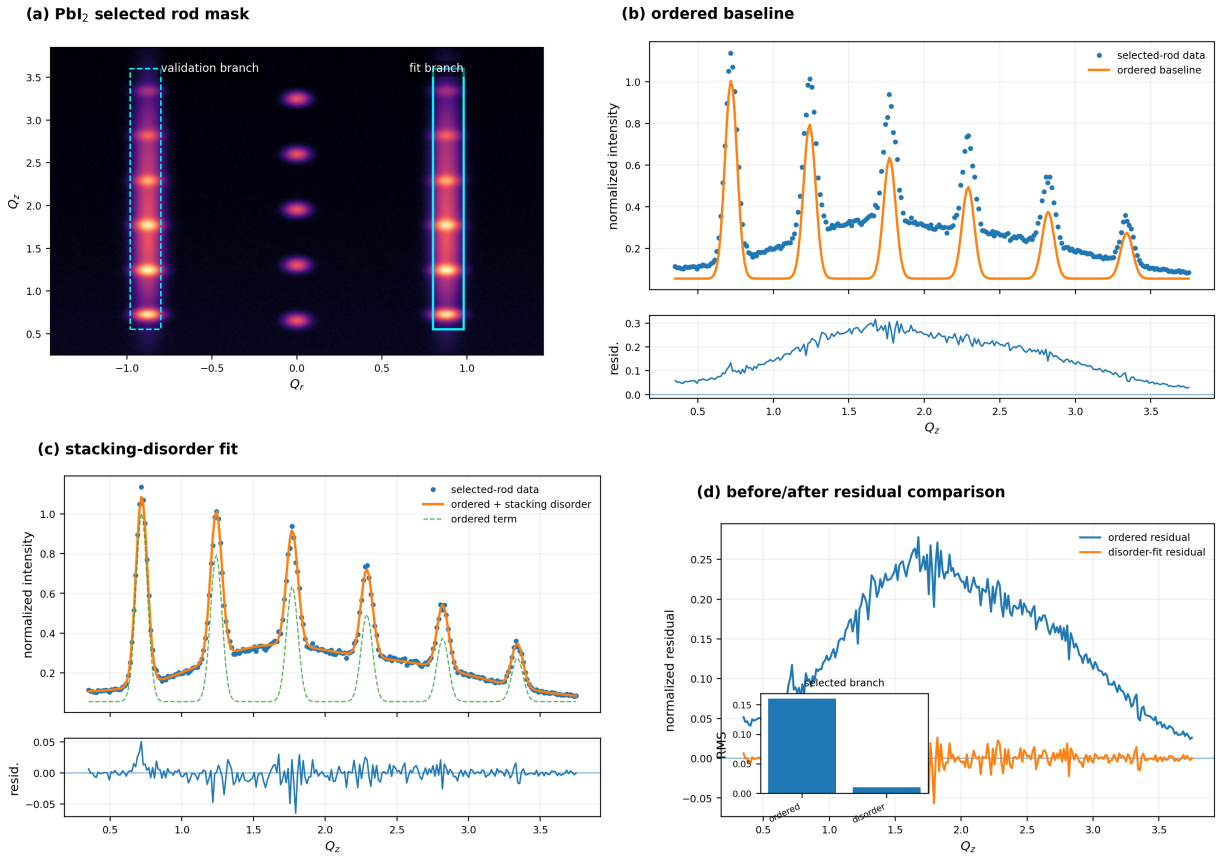

Figure 8: Selected-rod PbI_2 stacking-disorder refinement. (a) Caked detector-space view with the fitted rod mask and the opposite detector branch retained for validation. (b) Ordered-baseline profile and residual along the selected rod. The ordered model captures the principal Bragg positions but leaves diffuse between-peak intensity. (c) Selected-rod profile after adding the analytical stacking-disorder term while retaining the ordered baseline and upstream geometry/profile parameters. (d) Residual comparison before and after the disorder term is included.

Figure 8 shows the selected-rod comparison. The ordered baseline captured the principal Bragg locations but left systematic positive residual intensity along Q_z. Adding the analytical stacking-fault term reduced the selected-rod residual while leaving the upstream detector-space agreement intact. The fitted values were therefore interpreted as effective stacking-disorder descriptors for the rod and branch analyzed here.

6 Discussion and conclusion

The results show that 2D oriented-powder WAXS patterns can be analyzed quantitatively without reducing the image to either a single-crystal indexing problem or a conventional 3D powder pattern. The forward model keeps detector geometry, sample alignment, mosaic broadening, ordered structure, and stacking disorder within one detector-space calculation,

while the refinement separates these terms by the observables that constrain them most directly. Ordered Bi_2Te_3 and Bi_2Se_3 provide the baseline case where, once geometry and mosaicity are fixed, the remaining Bragg-intensity variation can be assigned to ordered-structure parameters. PbI_2 provides the disorder case. The ordered Bragg model reproduces the peak positions and intensities but leaves structured rod intensity that is reduced by adding the stacking-fault correction.

Peak positions determine geometry before peak widths are interpreted. Local profiles determine the mosaic and instrumental response before integrated intensities are assigned to structure. Selected Q_r/Q_z rod profiles determine stacking disorder only after the ordered baseline is fixed. That order reduces the risk that diffuse scattering is incorrectly absorbed into geometry, mosaicity, or independent peak areas.

The approach is also limited by the information retained in the selected observables. A detector-space forward model cannot remove ambiguity if the chosen reflections do not span enough geometry, mosaic, and intensity constraints. The stacking-disorder parameters are therefore reported as effective descriptors of the selected rod family and fitted model class. Within that scope, the combination of detector-space fitting and selected-rod validation provides a practical route for comparing ordered and faulted layered films using area-detector WAXS data.

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